

Digital-to-Analog Converters

June 2001

THE ANALOG DEVICES SOLUTIONS BULLETIN

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World's Smallest Direct Digital Synthesizer with Integrated DAC Dissipates Only 20 mW

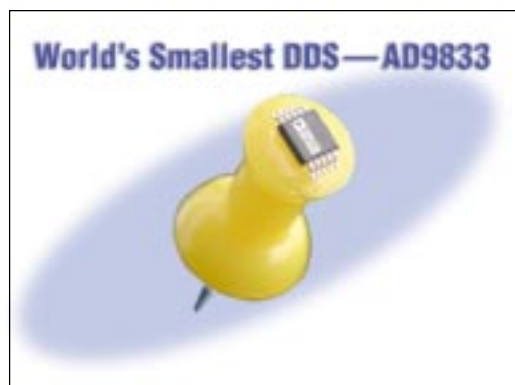
FREE OFFER!
SEE INSIDE!

Innovative direct digital synthesis solutions from Analog Devices have enabled many systems to “go digital” by performing the agile frequency synthesis function entirely in the digital domain. Now there's a new family of DDS products that reduces power consumption to levels that would have been unimaginable even a couple of years ago. The AD9833 and AD9834 consume only 20 mW of power while maintaining the excellent performance and feature-set associated with DDS products from Analog Devices. A power-down function that reduces the power consumption to less than 1 mW is available on both products. These products are also specified to operate on a single-supply voltage (2.5 V to 5.5 V). The AD9833 and AD9834 combine the numerical controlled oscillator, sine lookup table, frequency registers, phase registers, and a 10-bit D/A converter on a single chip.

	AD9833	AD9834
Clock Rate	25 MHz	50 MHz
DAC	10-Bit	10-Bit
SFDR	–75 dB min	–75 dB min
Power Cons.	20 mW typ.	25 mW typ.
Package	10-μSOIC	20-TSSOP
Price	\$3.50	\$4.95

The output of the AD9833 or AD9834 is a high-performance signal that can achieve a signal-to-noise ratio of –50 dB min and give a spurious free dynamic range of –75 dBc in a 50 kHz window. This level of performance, coupled with the ultrafine frequency tuning resolution, makes these products ideal for many applications.

With a 28-bit phase accumulator, frequency resolution can be controlled to 1 part in 250 million. With a 25 MHz clock this gives a resolution of 0.1 Hz. Two on-chip frequency registers and two on-chip phase registers enable preset frequency and phase words to be stored for later use. On the AD9834 the stored words in these registers can be enabled via a single external pin. The interface is a simple serial 3-wire interface with data being loaded by two 16-bit word writes.



APPLICATIONS

- portable instrumentation
- handheld communications systems
- RFID
- telemetry systems
- waveform synthesis



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www.analog.com/bulletins/dacs

Utilizing the Pin- and Software-Compatible Attribute of the AD53xx DAC Family

Many OEMs desire to provide product differentiation and/or cost-performance selection capability at the board assembly stage of their end system. They wish to achieve this by using a single PCB design that delivers selected levels of specified performance based on the installation of variable resolution/accuracy components. These key components must be completely interchangeable with respect to system hardware and software.



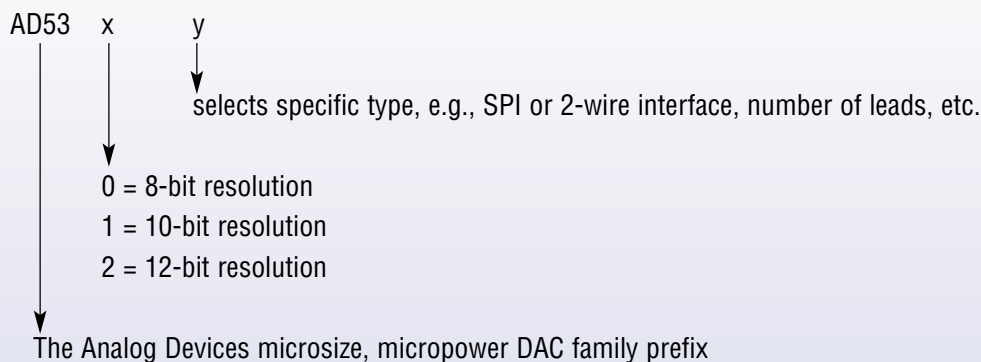
Pin- and software-compatible 8-/10-/12-bit, microsize, micropower AD53xx DACs from Analog Devices provide the mechanism for offering multiple levels of performance on a common PCB. The AD53xx family of DACs is the largest selection of the smallest DACs in the world. AD53xx DACs feature rail-to-rail voltage outputs, 100 μ A/50 nA operating/standby current, 2.5 V to 5.5 V single-supply operation. Single, dual, and quad parts are available with serial (SPI and 2-wire) and parallel interfaces. Available in SOT-23, μ SOIC, and TSSOP, these parts are particularly suitable for use in battery-powered, portable, and high-density equipment.

- multiple products from a single PCB design
- facilitates cost-performance selection at board assembly
- defers final product definition to late in the design/production process
- reduced inventory
- more flexible and robust design can compensate for changing feature requirements in product definition
- reduced design risk
- provides optimal cost vs. performance trade-off with no overhead cost

Hardware Design Process

Select pin-compatible serial interface 8-/10-/12-bit devices from the selection tables or trees. Every part is available in all three resolutions; each has rail-to-rail outputs and offers a wide supply voltage range. Design the system around the highest resolution 12-bit DAC, but allow for lower resolution devices to be used to meet correspondingly lower performance (and cost) end-system requirements. The only impact from interchanging DAC resolutions will be the accuracy performance; all other characteristics remain constant. This approach is a flexible and robust design solution for providing an end system that is scaleable in both performance and cost.

Part Selection:



For example, choose AD5300 for 8-bit, AD5310 for 10-bit, and AD5320 for 12-bit resolution voltage output DAC with serial SPI interface, and so on...

Software Design

Base the software design on the highest resolution part and make sure not to use any modes that depend on or are related to the number of bits of resolution.

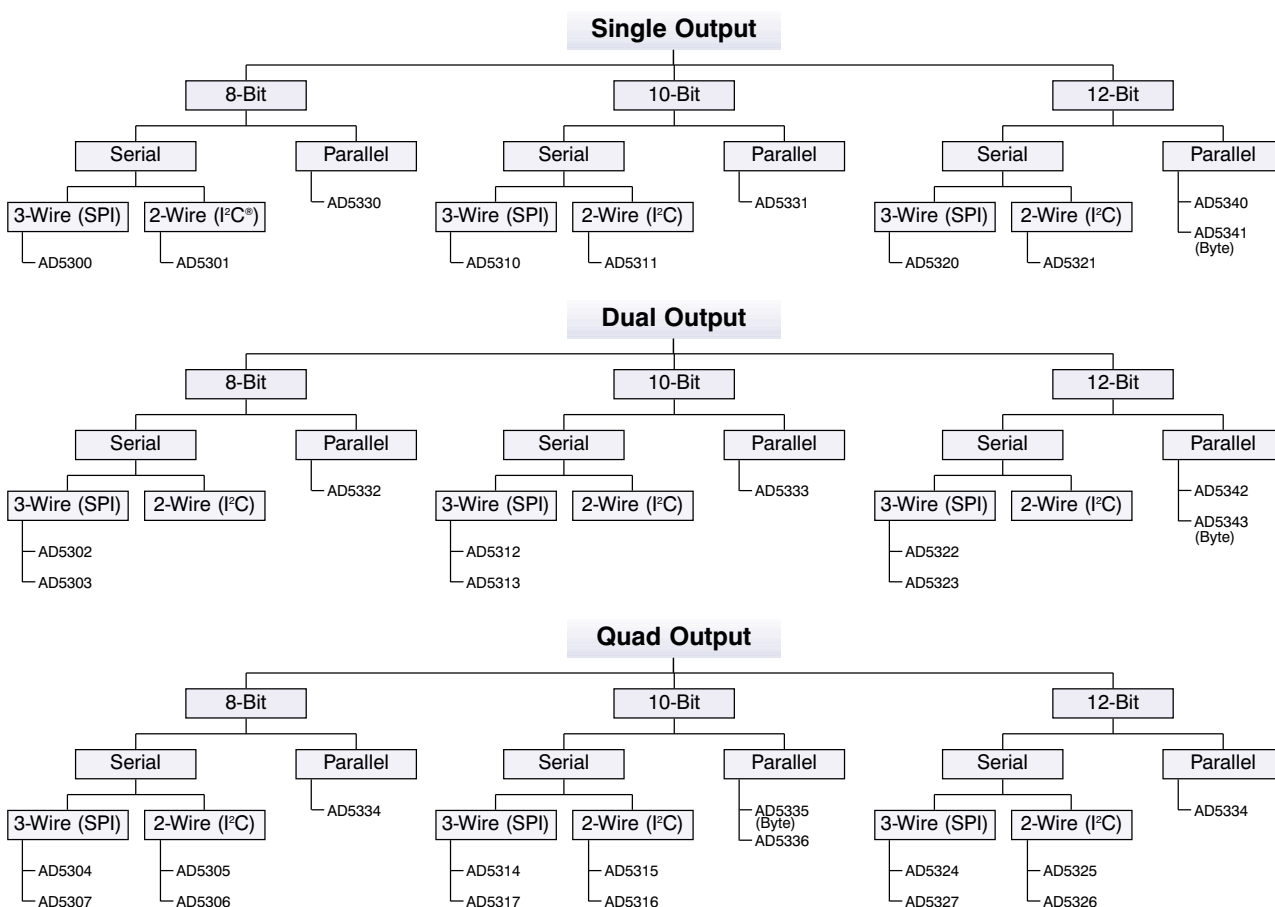
Example Input Register Contents

DB15 (MSB)	DAC data bits												DB0 (LSB)		Example Models		
X	X	PD1	PD0	D7	D6	D5	D4	D3	D2	D1	D0	X	X	X	X	8-Bit	AD5300
X	X	PD1	PD0	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	X	X	10-Bit	AD5310
X	X	PD1	PD0	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	12-Bit	AD5320

Notes:

1. Use the highest number of bits for data, i.e., 12 bits.
2. Use left-justified data, i.e., the MSB should be to the left. This will normally mean that the four most significant bits will be in a second byte also containing control bits, etc.
3. Set the lower (least significant) unused data bits to zero if the final design uses a resolution that is known to be permanently lower than originally designed for.

Selection Trees



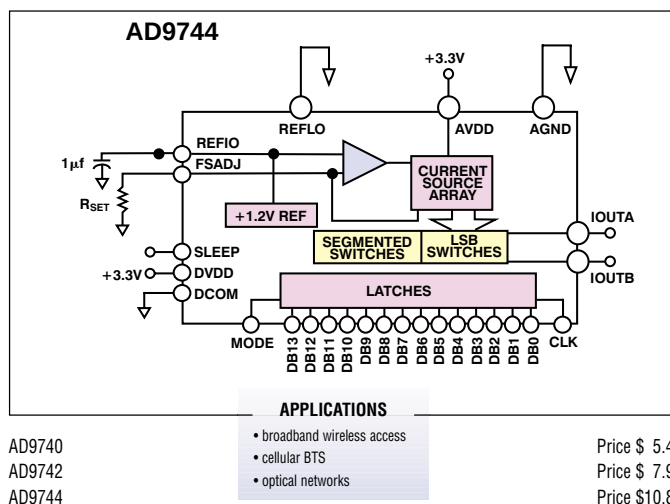
Third-Generation TxDAC Reaches New Heights in Superior Performance

ADI, the original inventors of the TxDAC, improves SFDR performance 5 dB – 7 dB over second-generation, with the AD9744, AD9742, and AD9740 Third-Generation TxDACs.

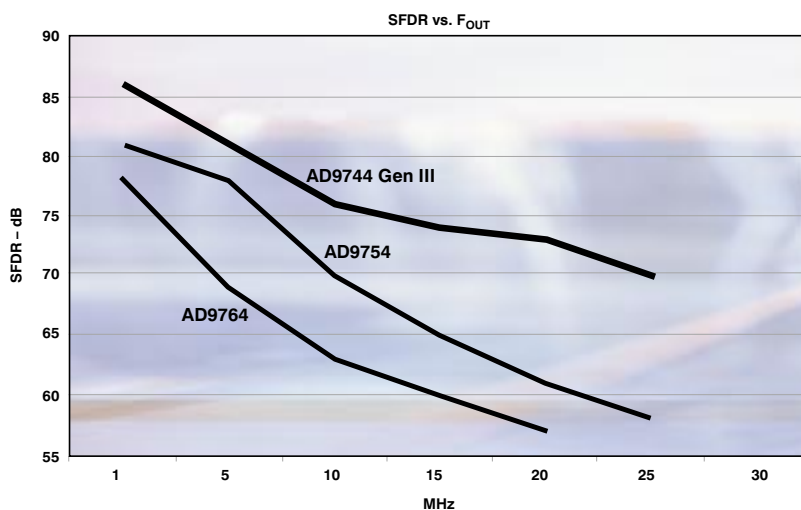
The world is rapidly evolving toward digital communications and digital architectures. Processing transmit signal data at “IF” rates requires high-speed D/A converters that are specially designed for maximum dynamic performance up to, and beyond, their Nyquist limit. When coupled with a need for low-power dissipation, this requirement for wideband dynamic performance limits the field of choices in D/A converters.



ADI's new family of Third-Generation TxDACs, the AD974x family, once again sets a new industry standard in both noise and distortion performance. The AD974x is a 165 MSPS, TxDAC industry-standard pin-compatible family, especially designed for +3.3 V low-power operation. This new family also incorporates a two's complement function for easy interfacing to ADI's high-performance SHARC® DSPs. With remarkably low power dissipation of less than 100 mW, up to 87 dB of spurious-free dynamic range (SFDR) can be achieved at 1 MHz output frequencies. Available in a 28-lead SOIC or TSSOP, these new DACs are ideal for wireless broadband, home networking, WLAN, and ultrasound applications. A new 32-lead, ultrasmall CSP package will be available by the end of 2001. Analog Devices is committed to delivering the world's best performance and value in TxDACs.



THIRD-GENERATION AD9744 SFDR PERFORMANCE VS. PREVIOUS GENERATIONS OF TxDACs



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Difficulties in Maintaining In-Band Noise Performance

In a 3G transmitter, maintaining in-band noise performance is critical in delivering the promised capacity and data rates. When processing a wideband channel, multiple IF stages with their respective mixers, filters, and gain blocks affect the noise figure of the transmitter. Each mixing stage has a noise figure impact and an insertion loss. To recoup the insertion loss, gain amplifiers must be utilized, which impacts the noise figure even further. Eliminating IF stages helps minimize system noise by reducing the number of active circuit elements. But the trade-off in achieving a single hop transmitter places severe performance burdens on the D/A converter.

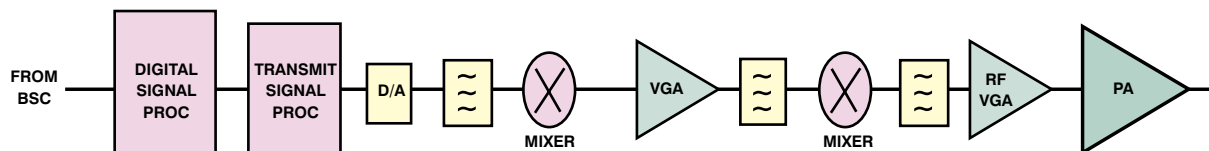


FIGURE 1: CLASSIC SUPER-HETERODYNE TRANSMITTER

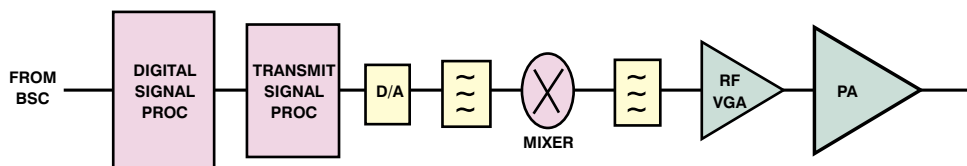
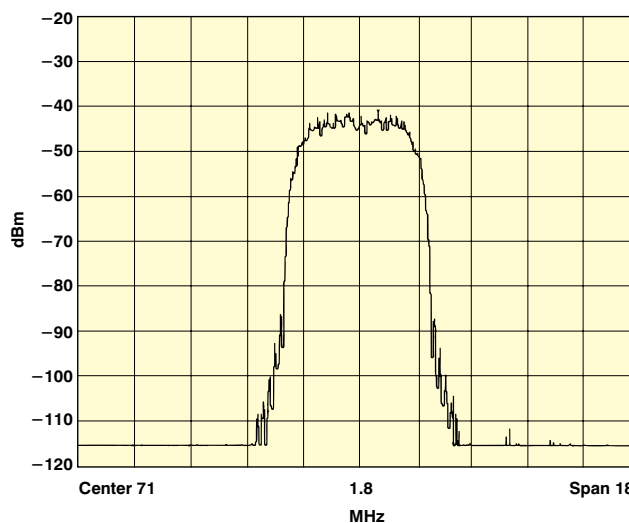


FIGURE 2: SINGLE HOP TRANSMITTER USING A DIRECT IF DAC



The AD9772A can now deliver excellent ACPR performance at 70 MHz IFs. With its dramatically improved noise floor, IMD, and PLL performance, it can now deliver 3 G/W-CDMA performance at IF. 70 dB ACPR is achieved when using the AD9772A for a 3 G/W-CDMA 5 MHz carrier. By deploying the internal $f_s/4$ digital modulator and taking advantage of the advanced 0.35 micron 400 MSPS DAC core, other IFs can be achieved from 50 MHz up to 120 MHz.



AD9772A

Price \$32.18

Digital-to-Analog Converters Product Information											
DIGITAL POTENTIOMETERS											
Part Number	Number of VRS per Package	Terminal Voltage Range	Interface Data Control	Nominal Resistance (K Ω)	Resolution (Number of Wiper Positions)	Power Supply (I _{DD})	Packages	Comments		Price \$ 100 pc US OEM	
AD5201	1	± 3 V, +5.5 V	3-Wire	10, 50	33	40 μ A	μ SOIC-10	Full AC Specs, Dual Supply, Pwr-On-Reset, Low Cost		0.75	
AD5220	1	+5.5 V	UP/DOWN	10, 50, 100	128	40 μ A	PDIP, SO-8, μ SOIC-8	No Rollover, Pwr-On-Reset		1.10	
AD7376	1	± 15 V, +28 V	3-Wire	10, 50, 100, 1000	128	100 μ A	PDIP-14, SOL-16, TSSOP-14	Single +28 V or Dual ± 15 V Operation, TC < 50 ppm/ $^{\circ}$ C		2.95	
AD5200	1	± 3 V, +5.5 V	3-Wire	10, 50	256	40 μ A	μ SOIC-10	Full AC Specs, Dual Supply, Pwr-On-Reset		1.00	
AD8400	1	+5.5 V	3-Wire	1, 10, 50, 100	256	5 μ A	SO-8	Full AC Specs		1.29	
AD5241	1	± 3 V, +5.5 V	2-Wire	10, 100, 1000	256	50 μ A	SO-14, TSSOP-14	I 2 C Compatible, TC < 50 ppm/ $^{\circ}$ C		1.10	
AD5231	1	± 3 V, +5.5 V	3-Wire	10, 50, 100	1024	20 μ A	TSSOP-16	Nonvolatile Memory, Direct Program, I/D, ± 6 dB Settability		2.20	
AD5260	1	± 5 V, +15 V	3-Wire	20, 50, 200	256	60 μ A	TSSOP-14	+5 V to +15 V or ± 5 V Operation, TC <50 ppm/ $^{\circ}$ C		2.05	
AD5222	2	± 3 V, +5.5 V	UP/DOWN	10, 50, 100, 1000	128	80 μ A	SO-14, TSSOP-14	No Rollover, Stereo, Pwr-On-Reset, TC <50 ppm/ $^{\circ}$ C		1.78	
AD8402	2	+5.5 V	3-Wire	1, 10, 50, 100	256	5 μ A	PDIP, SO-14, TSSOP-14	Full AC Specs, nA shutdown current		1.95	
AD5207	2	± 3 V, +5.5 V	3-Wire	10, 50, 100	256	40 μ A	TSSOP-14	Full AC Specs, Dual Supply, Pwr-On-Reset, SDO		1.39	
AD5232	2	± 3 V, +5.5 V	3-Wire	10, 50, 100	256	20 μ A	TSSOP-16	Nonvolatile Memory, Direct Program, I/D, ± 6 dB Settability		2.95	
AD5235	2	± 3 V, +5.5 V	3-Wire	25, 250	1024	20 μ A	TSSOP-16	Nonvolatile Memory, Direct Program, TC <50 ppm/ $^{\circ}$ C		5.63	
AD5242	2	± 3 V, +5.5 V	2-Wire	10, 100, 1000	256	50 μ A	SO-16, TSSOP-16	I 2 C Compatible, TC <50 ppm/ $^{\circ}$ C		1.95	
AD5262	2	± 5 V, +15 V	3-Wire	20, 50, 200	256	60 μ A	TSSOP-16	+5 V to +15 V or ± 5 V Operation, TC <50 ppm/ $^{\circ}$ C		2.88	
AD5203	4	+5.5 V	3-Wire	10, 100	64	5 μ A	PDIP, SOL-24, TSSOP-24	Full AC Specs, nA Shutdown Current		2.06	
AD5233	4	± 3 V, +5.5 V	3-Wire	10, 50, 100	64	20 μ A	TSSOP-16	Nonvolatile Memory, Direct Program, I/D, ± 6 dB Settability		3.95	
AD5204	4	± 3 V, +5.5 V	3-Wire	10, 50, 100	256	60 μ A	PDIP, SOL-24, TSSOP-24	Full AC Specs, Dual Supply, Pwr-On-Reset		2.23	
AD8403	4	+5.5 V	3-Wire	1, 10, 50, 100	256	5 μ A	PDIP, SOL-24, TSSOP-24	Full AC Specs, nA Shutdown Current		2.95	
AD5206	6	± 3 V, +5.5 V	3-Wire	10, 50, 100	256	60 μ A	PDIP, SOL-24, TSSOP-24	Full AC Specs, Dual Supply, Pwr-On-Reset		2.90	
PRECISION DACS											
Part	Resolution & Monotonic to		Channel Count	Output Ranges	Digital Interface	Integral Nonlinearity	Update Rate per Channel				

AD5582	12-Bit	4	+5 V to +15 V or ± 5 V	Parallel	± 1 LSB Max	250 kHz
AD5583	10-Bit	4	+5 V to +15 V or ± 5 V	Parallel	± 1 LSB Max	250 kHz
AD5532ABC-x	14-Bit	32	Adjustable ¹	Serial	$\pm 0.15\%$ Typ	475 kHz Max
AD5532BBC-1 ²	14-Bit	32	Adjustable ¹	Serial and Parallel	$\pm 0.006\%$ Typ ($\pm 0.018\%$ Max)	62 kHz Max ³
AD5533ABC-1 ²	14-Bit	32	Adjustable ¹	Serial and Parallel	$\pm 0.006\%$ Typ ($\pm 0.018\%$ Max)	50 kHz Max ³
AD5532HSABC	14-Bit	32	+5 V and ± 2.5 V	Serial	$\pm 0.10\%$ Typ	1.2 MHz Max
AD5516ABC-x	12-Bit	16	± 2.5 V, ± 5 V, ± 10 V	Serial	± 2 LSBs Max	50 kHz Max
AD5517ABC-x	14-Bit	16	± 2.5 V, ± 5 V, ± 10 V	Serial	± 8 LSBs Max ⁴	
AD7839AS	13-Bit	8	Adjustable within ± 10 V	Parallel	± 2 LSBs Max	33 kHz Typ
AD7841BS	14-Bit	8	Same as for AD7839	Parallel	± 2 LSBs Max ⁴	32 kHz Typ

HIGH-SPEED DACS

Generic	Resolution (Bits)	Update Rate (MSPS Min)	SFDR (dB) @ F_{out} (MHz)	Power Supply Voltage (V_{DD})	Power Dissipation	Description
AD9740	10-Bit	150	73 @ 40	Single (2.7 to 5)	65 mW	High-Performance TxDAC +
AD9742	12-Bit	150	77 @ 40	Single (2.7 to 5)	65 mW	High-Performance TxDAC +
AD9744	14-Bit	150	80 @ 40	Single (2.7 to 5)	65 mW	High-Performance TxDAC +
AD9772A	14-Bit	160	85 @ 5	Single (3.3 $\pm 6\%$)	215 mW	Improved $2\times$ Interpolating, Internal PLL, Direct IF, Wideband TxDAC +

DIRECT DIGITAL SYNTHESIZERS

Generic Number	Master Clock (MSPS Min)	Power Supply Voltage (V_{DD})	Power Dissipation (mW Max)	Description
AD9833	25	Single (2.5 V to 5.5 V)	33 mW	Ultralow Power, 50 MSPS Complete DDS with Integrated 10-Bit DAC, 10-Lead μ SOIC Package
AD9834	50	Single (2.5 V to 5.5 V)	33 mW	Ultralow Power, 50 MSPS Complete DDS with Integrated 10-Bit DAC, 20-Lead TSSOP Package
AD9852	300	Single (3.3 V)	1200	300 MSPS Complete DDS with 12-Bit DAC, High-Speed Comparator, Chirp Mode, Programmable REFCLK Multiplier
AD9854	300	Single (3.3 V)	1200	300 MSPS Complete DDS with Quadrature 12-bit DACs, High-Speed Comparator, Chirp Mode, Programmable REFCLK Multiplier

NOTES

¹Adjustable within a maximum span of 10 V for -1 , -3 , and -5 , and 20 V for -2 .

²The AD5532BBC-1 and AD5533ABC-1 are specified as Sample and Holds (with zero droop) and differ only in that the AD5532BBC-1 has reduced offset and gain.

³Based on the Sample and Hold mode of operation with a 16 ms acquisition time.

⁴14-bit LSBs.



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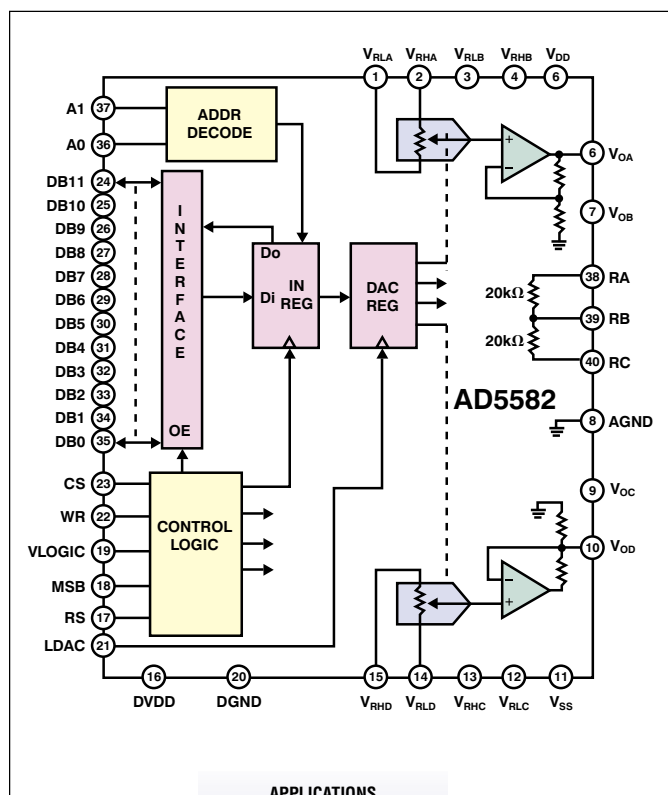
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First Multipurpose, Quad, 12-Bit DAC Provides Individual REF_H and REF_L for Each DAC

Many systems have voltage adjustment requirements around a very limited voltage span within the total power supply operation, and to further complicate the issue, these limited spans are different for each parameter. For example, in optical network control loops, various control voltages are necessary for extinction levels and average laser driver levels. In ATE equipment, unsymmetrical output voltage ranges (-3 V to $+7\text{ V}$) are necessary to test voltage levels on the IC device under test. The traditional costly approach is a full rail-to-rail, very high-resolution D/A converter.



The new AD5582/AD5583 offer quad 12-/10-bit digital-to-analog converters with separate reference inputs, allowing users to individually establish output spans with unsymmetrical ranges. These low-power (3 mA), fast settling time ($5\text{ }\mu\text{s}$) devices are offered in the compact TSSOP-48 package. Fast parallel digital interfacing allows 25 MHz data loading and read back. Operation from single $+5\text{ V}$ to $+15\text{ V}$ supplies as well as dual supply operation to $\pm 5\text{ V}$ ensures that most of the higher-voltage actuators can be directly controlled without further amplification to higher voltages. Ideal for instrumentation, industrial control, and optical network DWDM control circuits.



- quad DACs with separate reference inputs
- REF_H and REF_L enable unsymmetrical output spans
- data readback
- pin-compatible 12-bit and 10-bit devices
AD5582—12-bit
AD5583—10-bit
- compact 1 mm height PCMCIA-compatible TSSOP-48 package

AD5582
AD5583

APPLICATIONS

- optical network DWDM control circuits
- instrumentation
- industrial control systems

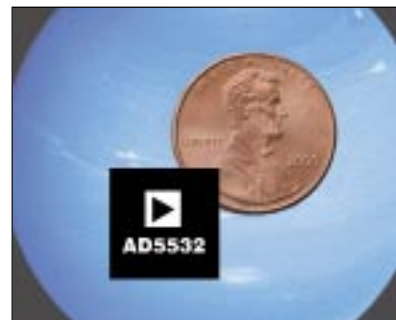
Price \$11.25
Price \$ 6.78

High-Density (8, 16, and 32 Channels) 12-Bit and 14-Bit DACs with Adjustable Output Voltage Ranges

Higher channel counts in optical communications, data acquisition, and industrial control applications are driving the demand for high-channel count D/A converters. The end-market applies constant pressure to reduce board space and system cost, and multichannel DACs are a means of increasing system capacities and performance with no sacrifice in system size and cost. However, the critical requirement is for multichannel precision DACs that combine high-channel count and superior accuracy.

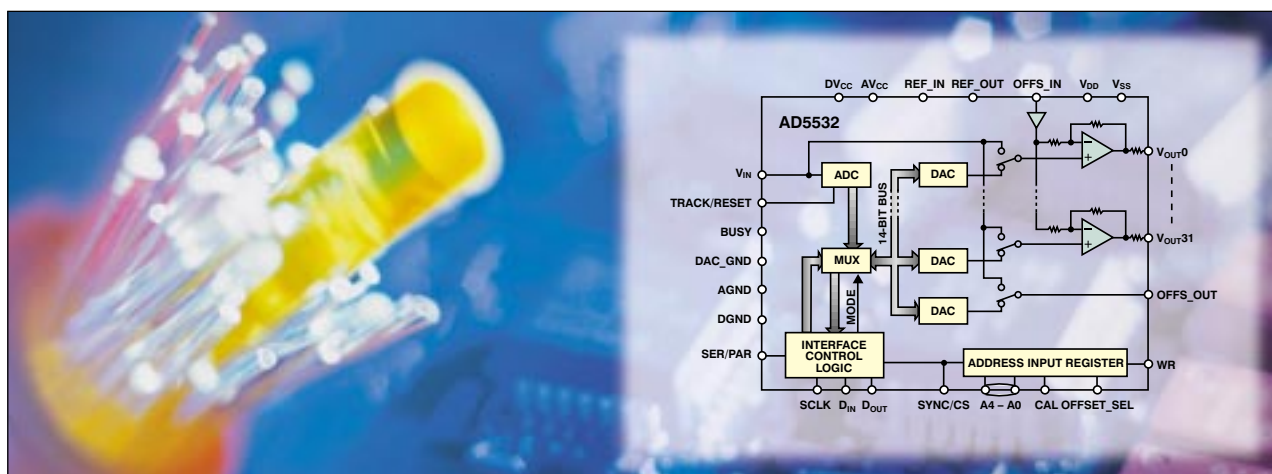
Solution

The flagship parts in this family of high channel-count DACs are the 32-channel, 14-bit AD5532s, which are packaged in 12×12 mm LFBGAs. The DACs are addressed via a 3-wire serial interface and DAC selection is accomplished through five address bits, A0 – A4. The per-channel update rate is 1.2 MHz for the AD5532HS, with a 30 MHz serial clock. Output voltage ranges are selectable from 0 V – 5 V up to ± 10 V. In the same LFBGA package are the 16-channel, hardware- and software-compatible, 12-bit and 14-bit AD5516/AD5517. On these parts, the serial interface operates at clock rates up to 20 MHz and, like the AD5532 models, is compatible with standard SPI, Microwire™, and DSPs. They come in three output voltage range options, ± 2.5 V, ± 5 V, and ± 10 V. All of these 16- and 32-channel DACs are monotonic to 14 bits (12 bits for the AD5516).



For the highest absolute accuracy requirements, the octal 14-bit and 13-bit resolution AD7841 and AD7839, in 44-lead PQFPs, offer accuracies as tight as ± 2 LSBs (@ 14 bits) of integral nonlinearity error in addition to 14-bit and 13-bit monotonicity. These are parallel loading devices and offer an adjustable output voltage range up to a maximum of ± 10 V.

These high channel-count precision DACs are ideal for level-setting applications in ATE, gain, and offset adjust in high-end electronic instrumentation, in power and temperature control loops in dense optical networking systems, and for precision measurement in micro-electromechanical all-optical switches.



AD5516
AD5517
AD5532
AD5533
AD7839
AD7841

Price \$26.85
Price \$31.00
Price \$49.95
Price \$43.75
Price \$27.50
Price \$28.65

(See page 7 for high-density DAC product information.)

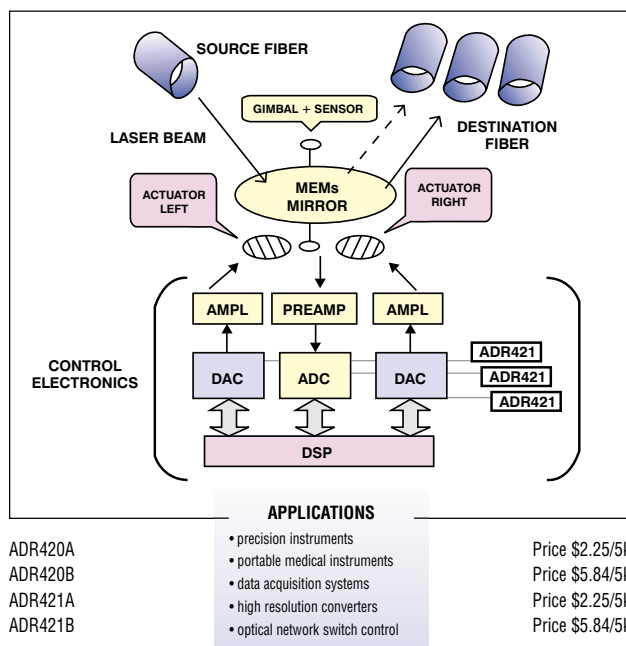
Ultraprecision Low-Noise XFET Voltage Reference

In the upcoming high-capacity, all-optical router network, provisioning of channel capacity employs arrays of micromirrors to direct and route optical signals from fiber to fiber, without first converting them to electrical form, which reduces the communication speed. The tiny micromechanical mirrors are positioned so that each is illuminated by a single beam, which carries unique information and can be passed to any desired input and output fiber. The mirrors are tilted by the dual-axis actuators controlled by precision ADCs and DACs within the control electronics. Due to the microscopic movement of the mirrors, not only the precision of the converters is important, the noise associated with these controlling converters is also extremely critical, because the number of converters employed can multiply total noise within the system.



The ADR421 voltage reference is perfect for these applications because its exceptionally low noise performance guarantees stability in the control loop. This device uses a new reference voltage generation technique known as XFET (eXtra implanted junction FET) which yields a reference with low noise, low supply current, and very low thermal hysteresis. The ADR42x XFET design provides $3\times$ improvement in system noise versus conventional bandgap references.

The ADR420/ADR421 are precision 2.048 V and 2.50 V XFET voltage references featuring low noise, high accuracy, high stability, and low power consumption in 8-lead SOIC and μ SOIC packages.



- initial accuracy: $\pm 0.04\%$ to $\pm 0.15\%$ max
- low noise: $1.75 \mu\text{V p-p}$
- low hysteresis: 40 ppm typical
- long-term stability: 50 ppm
- trim terminal: $\pm 0.5\%$ adjustment without affecting the tempco
- low tempco: 3 ppm/ $^{\circ}\text{C}$ and 10 ppm/ $^{\circ}\text{C}$ max

ADI Voltage References, Reference Them In Every Design:
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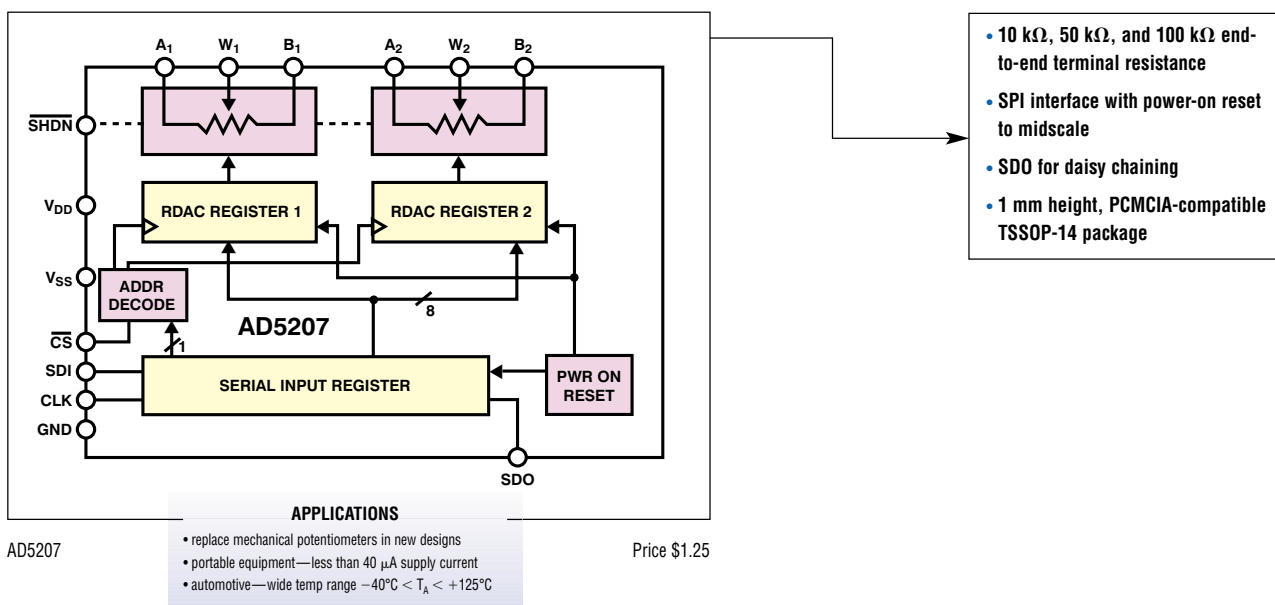
Compact 5×6 mm, Dual 8-Bit Digital Potentiometers

Electronic systems produced for high-volume markets demand increasingly lower production costs and cycle times. Manual adjustment, or “tweaking” of mechanical potentiometers as part of a final circuit calibration process is an expensive step that also slows down manufacturing throughput. In addition to being labor intensive, the mechanical potentiometer (albeit, a mainstay of electrical circuitry for the past 100 years) brings a host of other problems that affect system reliability. Mechanical wearout, sensitivity to vibration, the requirement for manual periodic re-adjustment, temperature sensitivity, aging effects, and the lack of processor-controlled functionality all combine to highlight the mechanical “pot” as a weak link in modern electronic system design.



These new 256-position digital potentiometers offer better than 1% accuracy in the position setting of resistance. Operating from single 3 V to 5 V supplies at less than 40 μ A, these digital potentiometers are ideal for portable applications. In addition, for ground-referenced AC signal adjustment, the AD5207 can be operated from dual ± 2.7 V supplies to maintain dc-coupled signal paths with low glitch noise. An override shutdown feature disables the POT for immediate power save in the resistor path. Full analog specifications of resistor tolerance, temperature coefficient, linearity, bandwidth, noise, distortion, and settling time assure the discriminating user of analog circuits a successful design.

The AD5207 is ideal for users needing to program resistance, current, or voltage in all types of electronic equipment.



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ANALOG DEVICES' SOLUTIONS BULLETINS AT A GLANCE

ADCs
DACs
Amplifiers
Communications
Power Management
High-Speed Linear

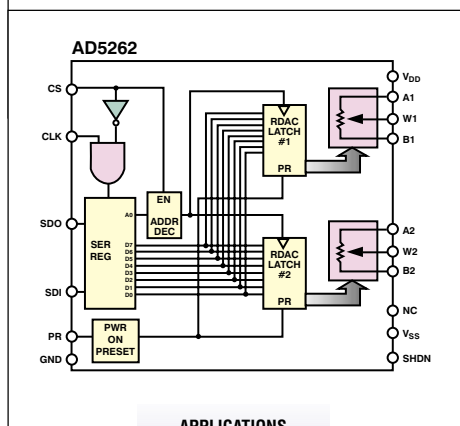
Wide Operating Voltage Range Digital Potentiometer for Wide Dynamic Range in Signal Adjustment

The need for processor control and adjustment in electronic systems has quickly led to the prevalence of the digital potentiometer. The advantages to digital potentiometers are compelling: vibration immunity, no manual "tweaking" required, elimination of aging effects, to name a few. However, some signal adjustment range is wider than what can be accommodated by typical digital potentiometers. Voltage spans of ± 15 V or ± 5 V are frequently required in industrial control applications.



These new digital potentiometers offer very wide terminal voltage operation for the most demanding signal adjustments. Either full ± 15 V single supply range signals, or an optional ± 5 V dual range can be accommodated for ground-referenced ac signals. Controlled from a 3-wire SPI-compatible data interface, the AD5260/AD5262 single/dual range digital POTs perform the same adjustments as potentiometers, trimmers, and variable resistors. The AD5260/AD5262 contain

the SPI logic interface, and level shifters, to operate the 3-terminal digital potentiometer settings through 256 steps.



APPLICATIONS

AD5260
AD5262

- setting 12 V actuators
- liquid crystal display control

Price \$1.85
Price \$2.60

- 20 k Ω , 50 k Ω , and 200 k Ω end-to-end terminal resistance
- SPI interface
- low absolute resistor TC < 50 ppm/ $^{\circ}$ C

*USD 1,000s (unless otherwise noted), recommended resale, FOB U.S.A.
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